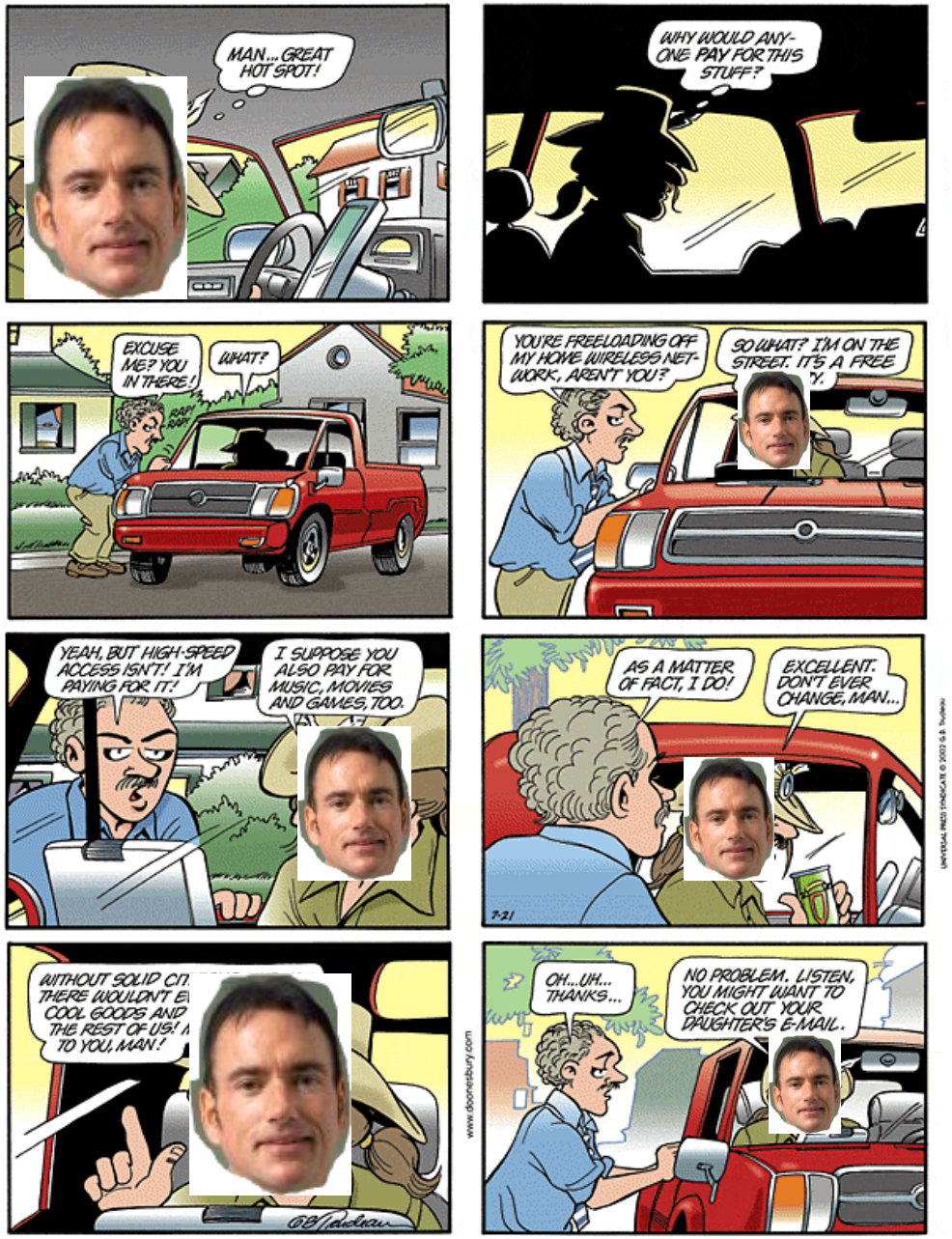
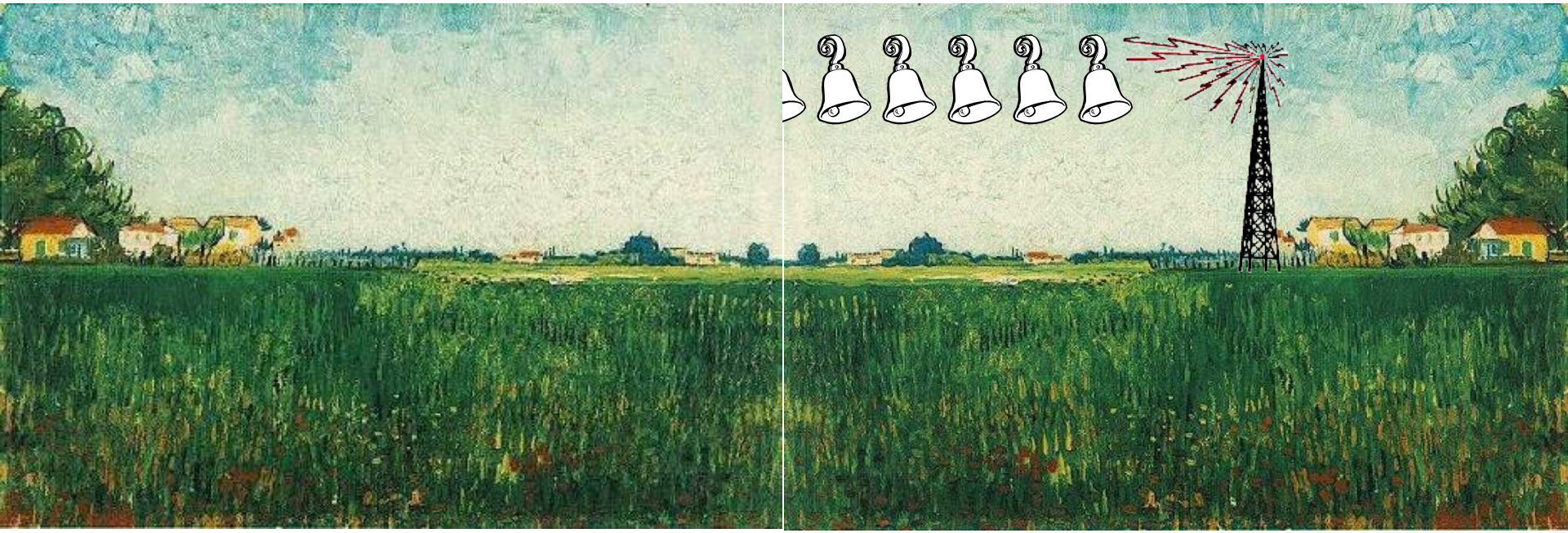
**You Didn’t Just Read This.**

# Pay Attention Jonathan!



# Bell Tolls

## *the quest for the bandwidth of the quantum*

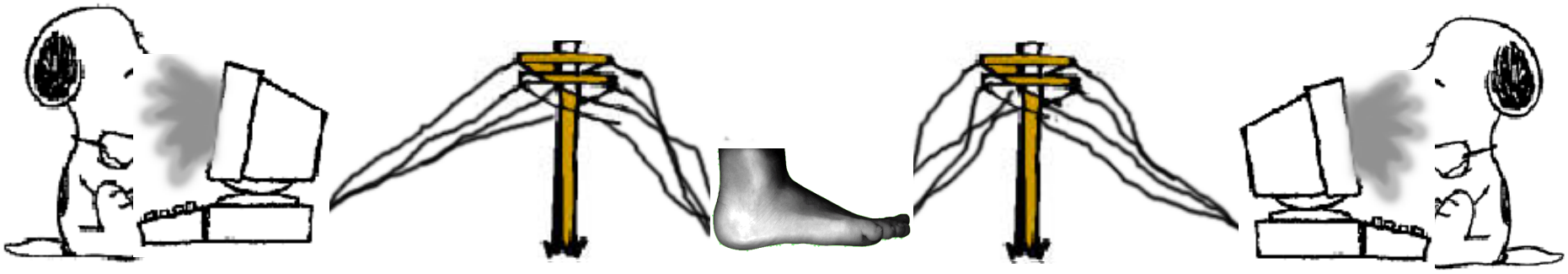



**Dave Bacon**

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California Institute of Technology

# Communication Breakdown



1. Initial unlimited classical communication between computers.
2. Communication lines cuts.
3. Memory erased except for data collected prior to 2.
4. Input into each computer.
5. Output by each computer.

Repeat

Build up table of input-output correlations

What sort of correlations can be produced?

# Bell Inequalities



Bell inequalities yield restrictions on correlations.

$$|\langle A_1 B_1\rangle + \langle A_1 B_2\rangle + \langle A_2 B_1\rangle - \langle A_2 B_2\rangle| \leq 2$$

$A_i$, $B_i$ are $\pm 1$ valued observables

Bell inequalities answer an information theoretic question:

What correlations can be produced using shared randomness and no communication between parties?

Make replacement

1. Initial unlimited quantum communication between computers.

   $\rightarrow$ parties can share entangled states.

$$|\langle \mathbf{A_1 B_1}\rangle + \langle \mathbf{A_1 B_2}\rangle + \langle \mathbf{A_2 B_1}\rangle - \langle \mathbf{A_2 B_2}\rangle| = 2\sqrt{2}$$

# Capitalism for Correlations

Shared randomness
Communication
Quantum correlations
Super-quantum correlations



Shared randomness
No communication
Classical correlations

Bell inequalities

How much communication is necesary to produce quantum correlations?

What are the COSTS of producing quantum correlations?



# What Can We Do?



1. Initial unlimited classical communication between computers.
2. Communication lines cuts.
3. Memory erased except for data collected prior to 2.
4. Input into each computer.

4.5. Metered rounds of communication

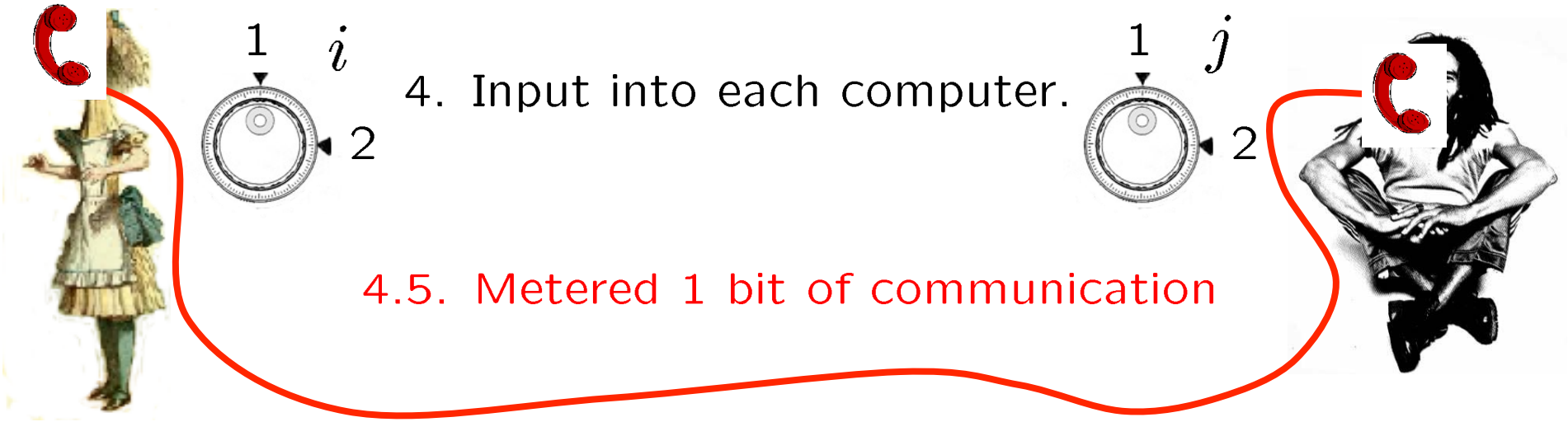
5. Output by each computer.

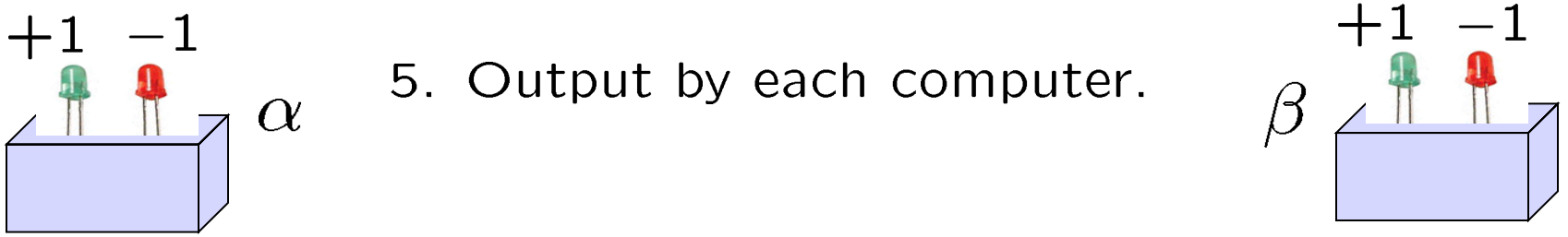
Repeat

What correlations can be produced using ~~$X$ amount~~ 1 bit of communication?

# Scenario EZ

Shared randomness + 1 bit of communication

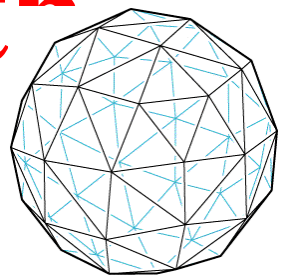



Probability of getting outcomes $i,j$ given input $\alpha$, $\beta$:

$\Pr(\alpha, \beta | i, j)$ 16 numbers.

# Bell Inequalities with Communication



Region of accessible probabilities is a polytope.

Faces of the polytope are necessary and sufficient set of Bell Inequalities with Communication for Scenario EZ.

Example inequality:

$$\Pr(+1,+1|1,1)+\Pr(-1,+1|1,2)+\Pr(+1,-1|2,1)+\Pr(-1,-1|2,2) \leq 2$$

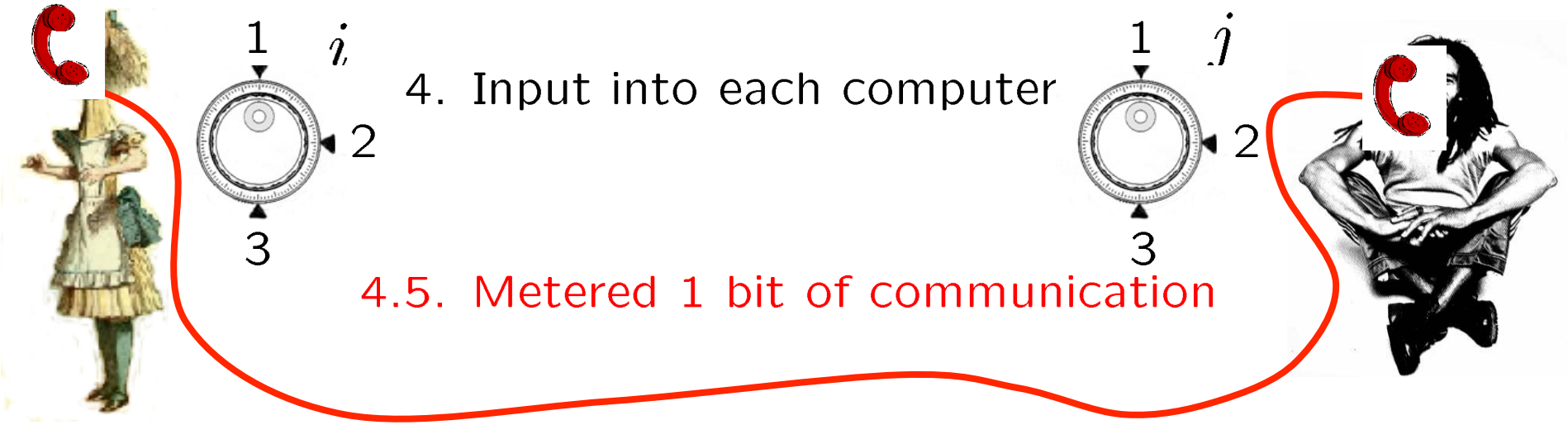
Does quantum theory satisfy these inequalities?

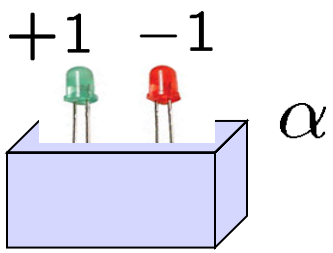
Any theory which produces correlations which do not allow communication by changing input satisfies all the inequalities.

All two party quantum measurement scenarios where each party selects one of two possible two outcome measurements can be produced by shared randomness and a one bit of communication.

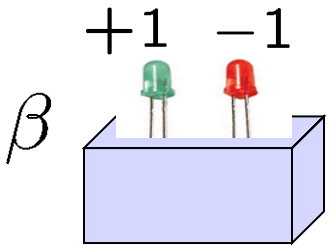
# Three Choices

Shared randomness + 1 bit of communication




5. Output by each computer.




Interested in the parties correctly producing the joint expectation value given measurements $i$ and $j$:

$$\langle \alpha_i \beta_j \rangle$$

# Bell Observable Inequalities with Communication

Example inequality:

$$-\langle\alpha_1\beta_2\rangle + \langle\alpha_1\beta_3\rangle - \langle\alpha_2\beta_1\rangle + \langle\alpha_2\beta_2\rangle + \langle\alpha_2\beta_2\rangle + \langle\alpha_3\beta_1\rangle + \langle\alpha_3\beta_2\rangle + \langle\alpha_3\beta_3\rangle \leq 6$$

All inequalities (nec. and suf. for joint observable):

$$\sum_{i,j=1}^{3} M_{ij}\langle\alpha_i\beta_j\rangle \leq 1$$

$$M = \frac{1}{6}\begin{bmatrix} 0 & -1 & 1 \\ -1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \quad \blacksquare \quad M = \frac{1}{11}\begin{bmatrix} 1 & 2 & -2 \\ 2 & 1 & 2 \\ -2 & 2 & 1 \end{bmatrix}$$

+ all matrices obtained by permuting rows or columns and multiplying rows or columns by $\pm 1$.

These inequalities are necessary but not sufficient for the full probabilities $\Pr(\alpha, \beta | i, j)$.

# Anything You Can Do, I Can Do Better

Quantum case:

$$\sum_{i,j=1}^{3} M_{ij}\langle \mathbf{A}_i \otimes \mathbf{B}_j \rangle \leq 1 \qquad \mathbf{A}_i^2 = \mathbf{B}_j^2 = \mathbf{I}$$

By examining $\left(\sum_{i,j} M_{i,j}\mathbf{A}_i \otimes \mathbf{B}_j\right)^2$ it follows that all inequalities are satisfied in the quantum case.

All two party quantum measurement scenarios where each party selects one of three possible two outcome projective measurements and the goal is to produce the joint correlation observable can be produced by shared randomness and a one bit of communication.




# Morals




- To understand quantum correlations we must move beyond the communication/no communication divide provided by Bell inequalities.
- Bell inequalities with communication provide bounds on correlations produced by augmenting shared randomness with a set amount of communication.
- For the few measurement scenarios, quantum correlations can be produced using a single bit of communication.
- Nonlocal hidden variable theories are important for understanding the classical/quantum computation gap?

# Fin




Collaborators

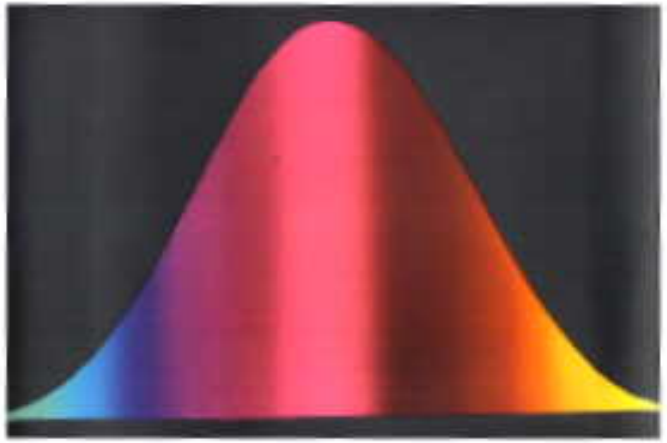
**Ben Toner**, Caltech

**Allison Coates**, U.C. Berkeley



# Title

$\begin{pmatrix} 1 \\ 0 \end{pmatrix}$





4. Output by each computer.





